Lab 2

Hardware Interfacing, Filtering, and Modelling

Responsible Lab TA: Adam Bigsby

Group 05

Team members:
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Team Contribution Table:

Team member	Experimental work and data collection	Analysis	MATLAB/ Simulink Code	Interpretation and Validation	Report Writing and Editing	Overall Contribution
Member 1 (Saud Amjad):	**S.A.**	**S.A.**	**S.A.**	**S.A.**	**S.A.**	33.34%
Member 2 (Hamzah Nadeem):	**H.N**	**H.N**	**H.N**	**H.N**	**H.N**	33.33%
Member 3 (Temuulen Ankhbayar):	**T.A**	**T.A**	**T.A**	**T.A**	**T.A**	33.33%
TOTAL:						100%

Acknowledgement: By submitting this report, all team members confirm that the information above accurately represents their contributions and that all team members have reviewed and approved the final report and associated files

6.2: Reading the Encoder

The conversion factor we used to convert the HIL reading to degrees was to multiply by a factor of 360/2048. Dividing the HIL reading by 2048 gives a fraction of how much the disk has spun, and by multiplying that fraction by 360, we can get the degrees rotated. Below is a block diagram of the implementation of the conversion factor:

$$\frac{(HIL\ Read)_{counter}}{2048} \times 360° = (HIL\ Read)_{degrees}$$

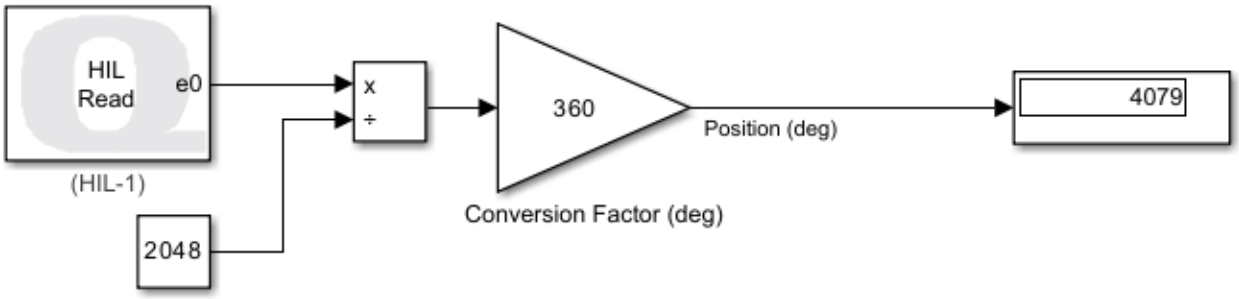


Figure 01: Conversion Factor to Convert Count Read to Degrees

If the angle was to be shown in radians, the multiplication of 360 would be changed to 2π.

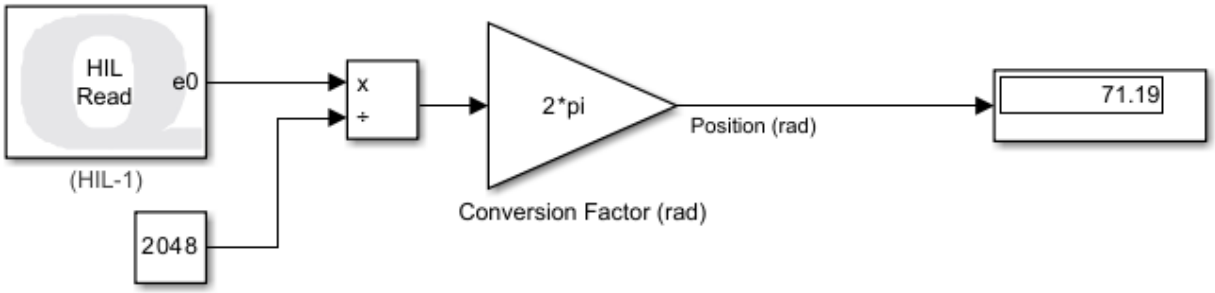


Figure 02: Conversion Factor to Convert Count Read to Radians

6.4: Filtering

2. From the figure shown below, we can see the initial angular speed climb from 0 to ~25 rad/s when the input voltage is 1V. As the input voltage increases from 1V to 3V, the angular speed rises to a value of ~70 rad/s. When the input voltage decreases from 3V to 1V, the angular speed falls to ~25 rad/s. This output speed trend continues as the square wave input changes. As this signal is unfiltered, we can see there is noise in the output angular speed signal.

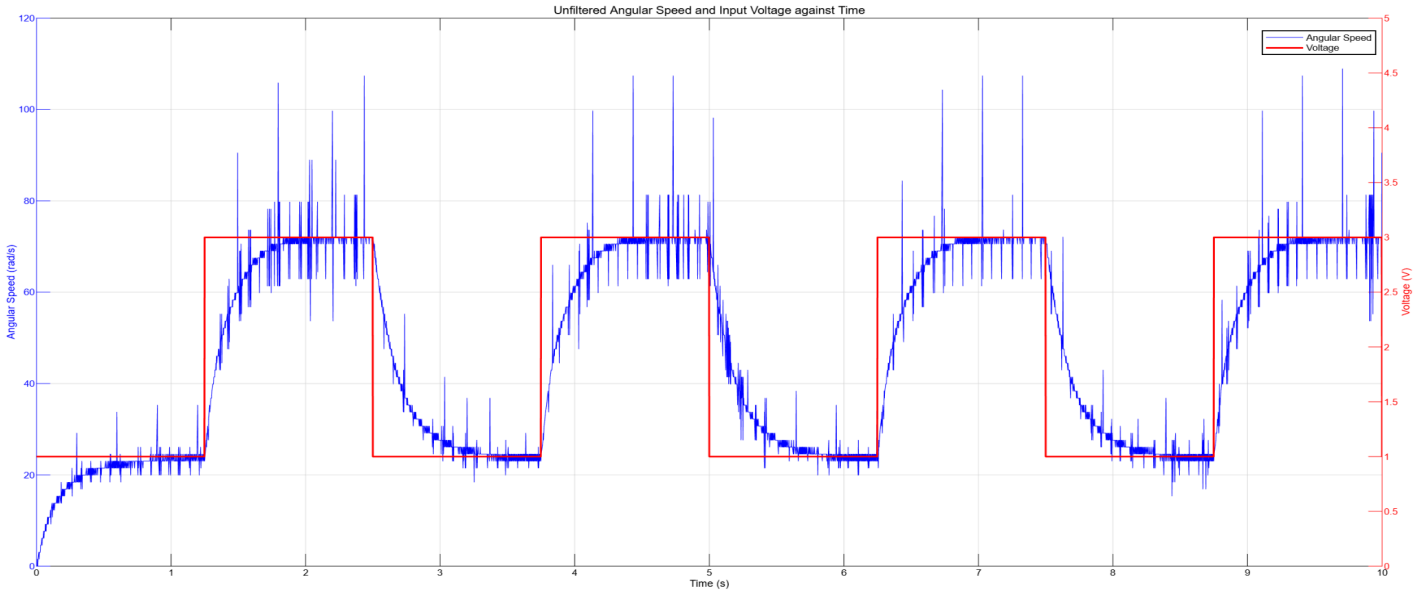


Figure 03: Unfiltered Angular Speed and Input Voltage against Time

3. The encoder-based measurement is noisy because the reader measures the results in steps of 0.002 seconds. Because of the step it reads at, the position graph is not a smooth curve, but instead a step function that changes value every 0.002 seconds, as shown below in Figure 04. When the position increments every 0.002 seconds, the derivative of the step increase will result in extremely large values on the angular speed plot. This is observed as the noise in the angular speed plot in Figure 03.

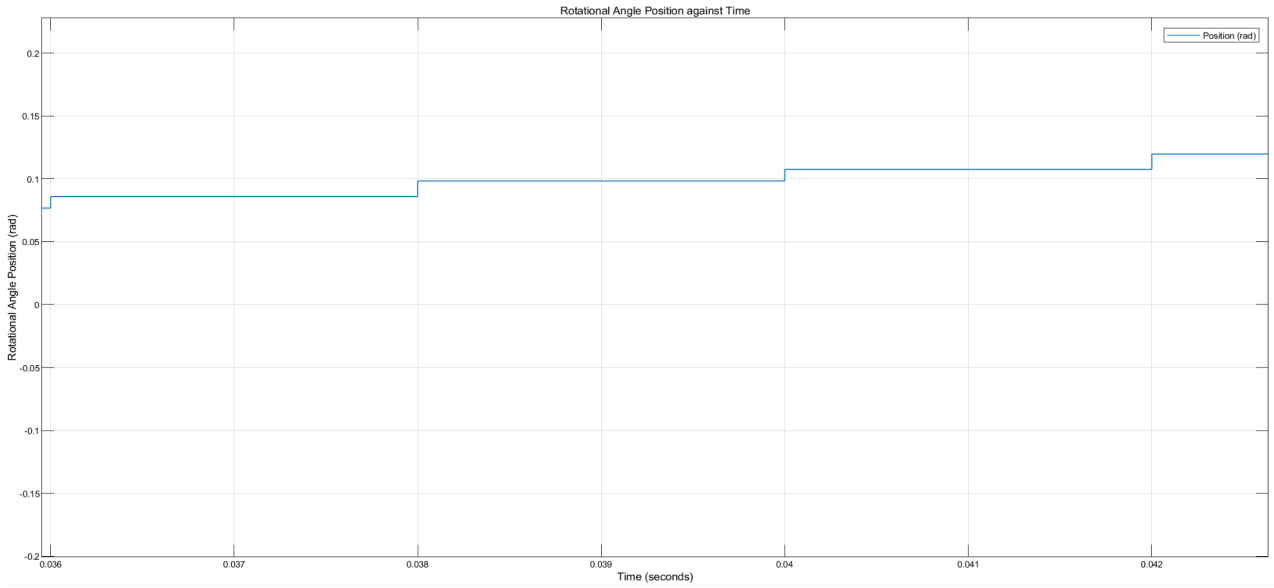


Figure 04: Zoomed In Rotational Angle Position (rad) Against Time

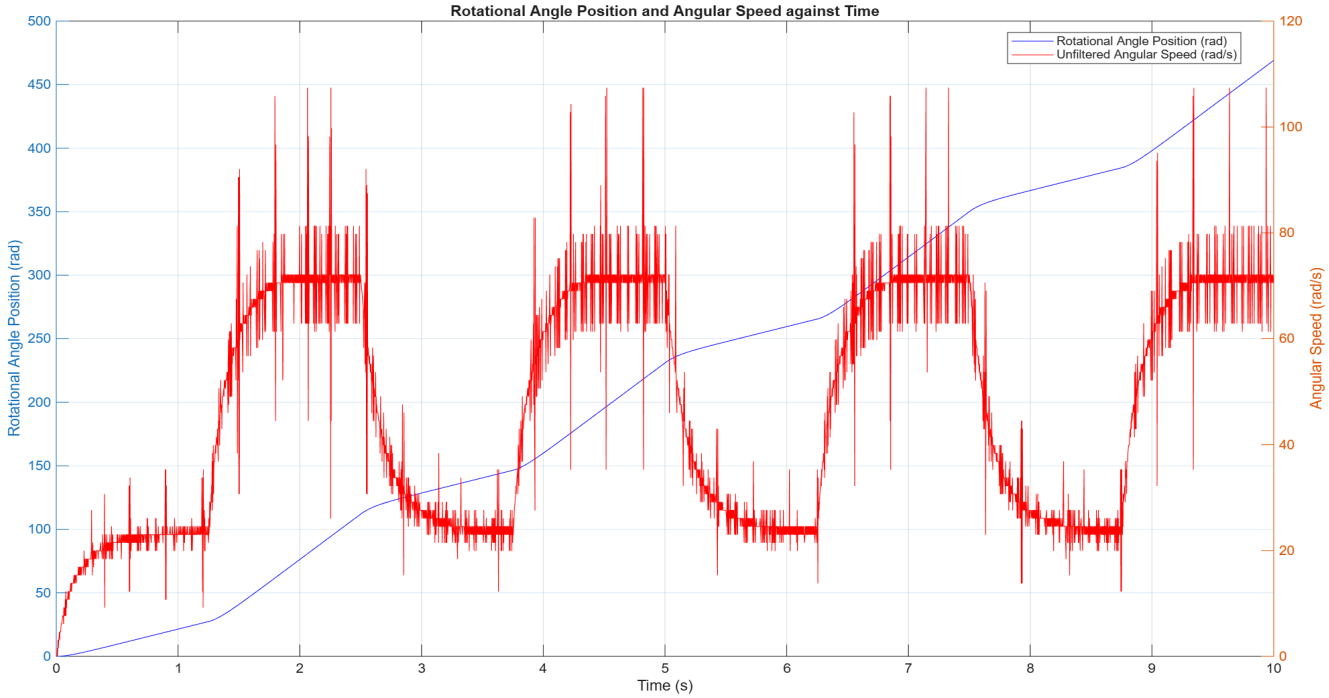


Figure 05: Angular Speed (rad/s) and Rotational Angle Position (rad) Against Time (Before Filtering)

5. As can be seen below, the output angular speed, after implementing the low-pass filter, has now greatly reduced in noise. By comparing this with Figure 05, it can be seen that there are less fluctuations

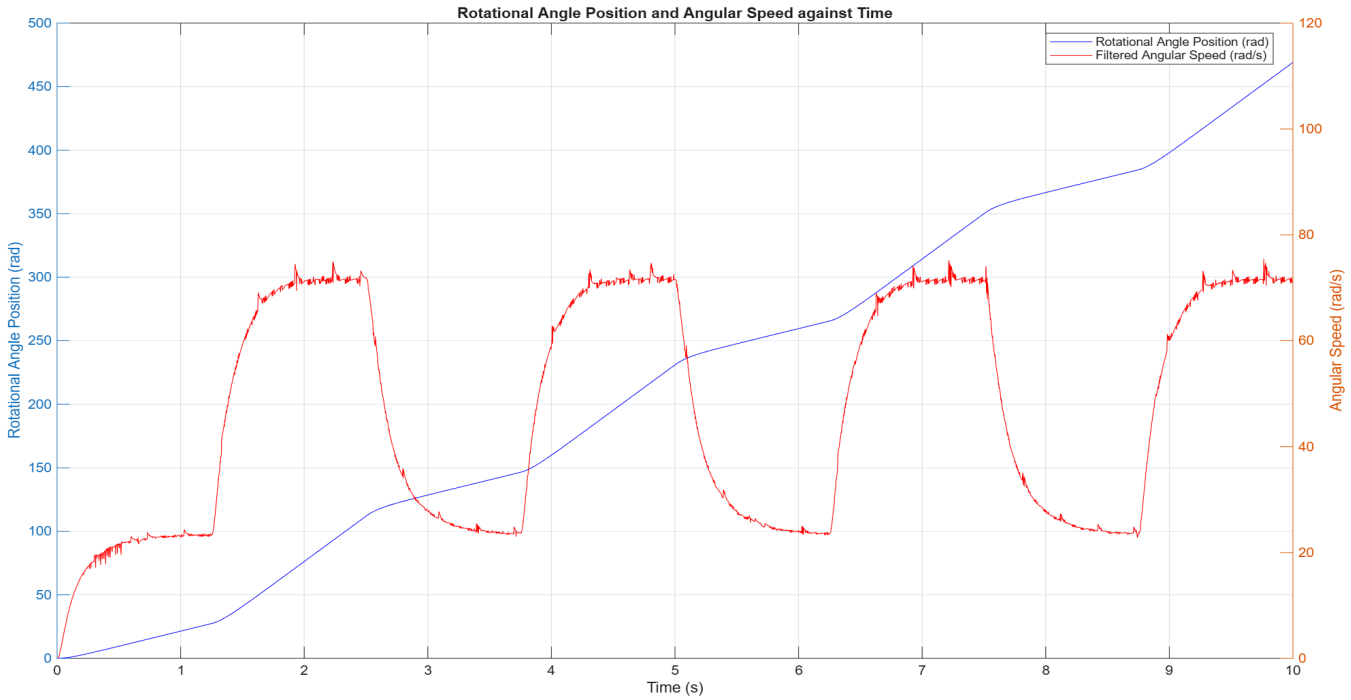


Figure 06: Angular Position and Filtered Angular Speed Against Time

6. The transfer function of a first order low-pass filter is defined:

$$G(s) = \frac{\omega_f}{s+\omega_f},$$

where ω_f is the cut-off frequency of the filter in radians per second ($\frac{rad}{s}$). We can then use the relationship $\omega_f = 2\pi f_c$, where f_C is the cutoff frequency in Hz.

We are provided the transfer function for the low-pass filter:

$$G(s) = \frac{(50)}{s+(50)},$$

where the cut-off frequency is **50** $\frac{rad}{s}$. Using the relationship $\omega_f = 2\pi f_c$, the cutoff frequency, f_C, in Hz is **7.96 Hz**.

6.5: Modelling using Differential Equations: - MATLAB Live Script (.mlx) file attached on D2L

The following is the derivation for the governing differential equations. From the lab manual, the following equations are already defined:

$$\tau_m = k_t i_m(t),\ \tau_m(t) = J_{eq}\frac{d\omega_m(t)}{dt}$$

We can make the substitution for τ_m and then isolate for $i_m(t)$:

$$i_m(t) = \frac{1}{k_t} \times \tau_m = \frac{J_{eq}}{k_t}\frac{d\omega_m(t)}{dt}$$

The lab manual also shows the equation from Kirchhoff's Voltage Law and then isolate for $i_m(t)$:

$$v_m(t) - R_m i_m(t) - k_m \omega_m(t) = 0$$

$$i_m(t) = \frac{1}{R_m}[v_m(t) - k_m \omega_m(t)]$$

We can then make the substitution for $i_m(t)$ and isolate for $\frac{d\omega_m(t)}{dt}$:

$$\frac{J_{eq}}{k_t}\frac{d\omega_m(t)}{dt} = \frac{1}{R_m}[v_m(t) - k_m \omega_m(t)]$$

$$\frac{d\omega_m(t)}{dt} = \frac{k_t}{R_m \times J_{eq}}[v_m(t) - k_m \omega_m(t)]$$

We can also use the following equation for the angular position (θ):

$$\frac{d\theta_m(t)}{dt} = \omega_m(t)$$

This makes the final set of differential equations:

$$\frac{d\omega_m(t)}{dt} = \frac{k_t}{R_m \times J_{eq}}[v_m(t) - k_m \omega_m(t)]$$

$$\frac{d\theta_m(t)}{dt} = \omega_m(t)$$

Many of the parameters are defined in the lab manual including:

$$R_m = 8.4\Omega,\ k_t = 0.042\ N.m/A,\ k_m = 0.042\ V/(rad/s),$$

$$J_m = (4.0) * 10^{-6} kg.m^2,\ J_h = (0.6) * 10^{-6} kg.m^2,\ m_d = 0.053\ kg,\ r_d = 0.0248\ m$$

We can determine J_{eq} using the following equation:

$$J_{eq} = J_m + J_h + J_d$$

$$J_{eq} = J_m + J_h + \frac{1}{2}mr^2 = ((4.0) * 10^{-6}) + ((0.6) * 10^{-6}) + \frac{1}{2}(0.053)(0.0248)^2$$

$$J_{eq} = 2.089856 \times 10^{-5}\ kg.m^2$$

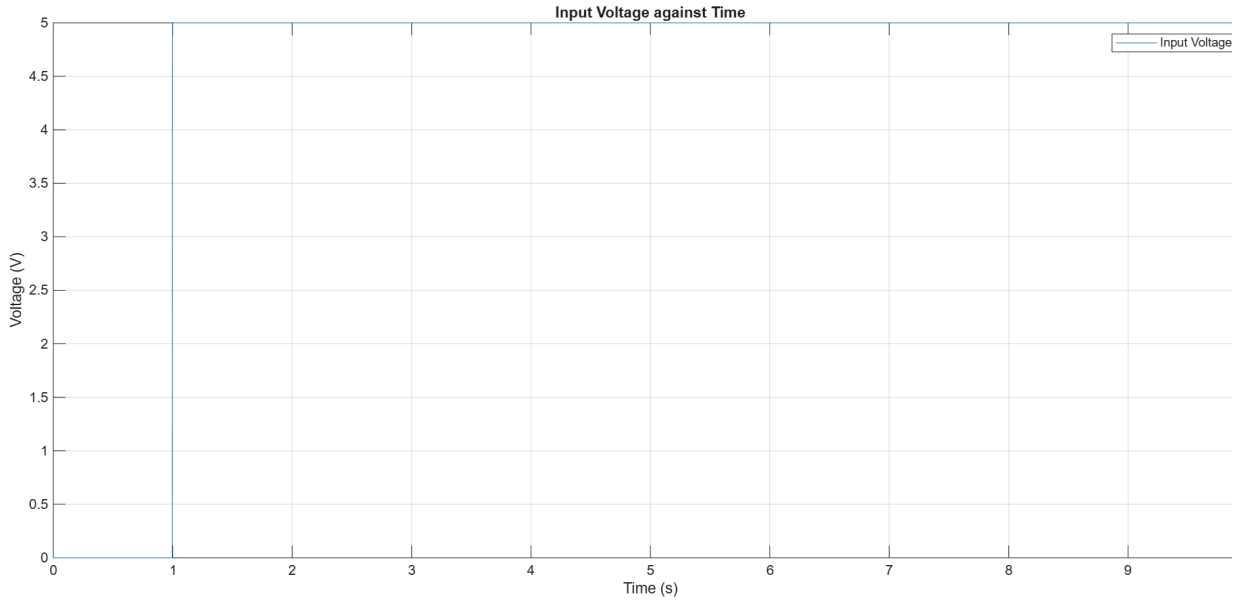


Figure 07: Input Voltage Plot Against Time for Section 6.5

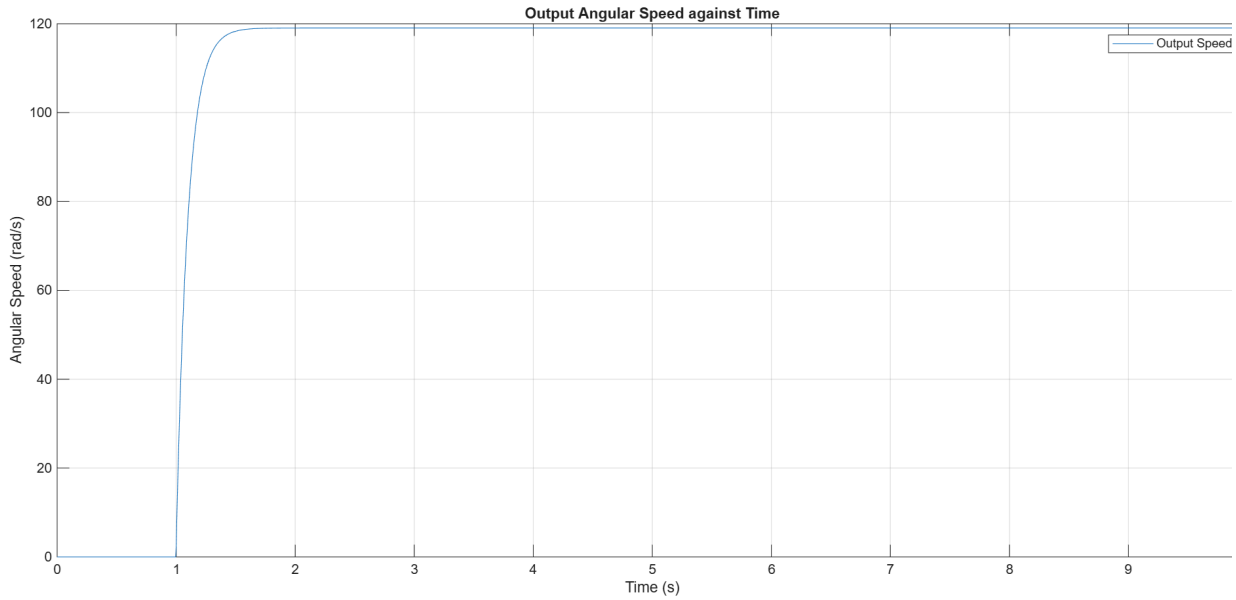


Figure 08: Output Angular Speed (rad/s) Against Time for Section 6.5

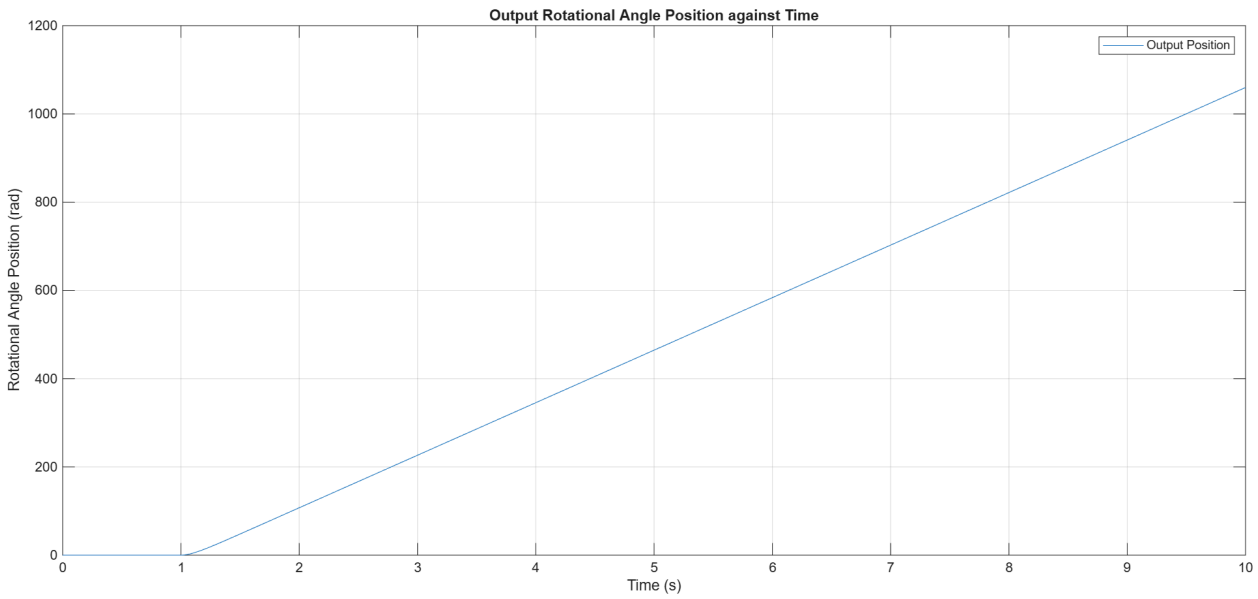


Figure 09: Output Rotational Angular Position (rad) Against Time for Section 6.5

6.6: Modeling using Block Diagrams: - Simulink Model File (.slx) file attached on D2L

1. The following is the block diagram model of the QUBE-Servo-2 developed in Simulink:

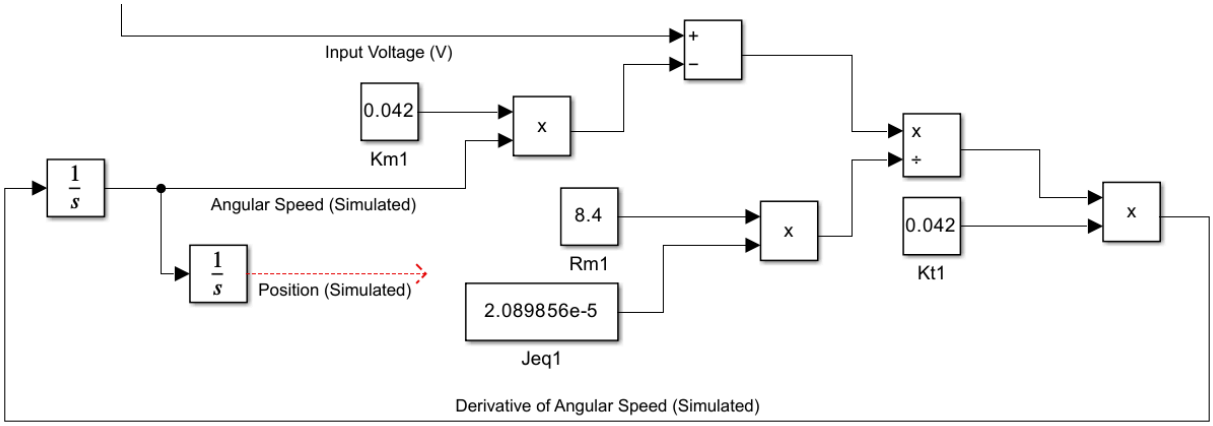


Figure 10: Block Diagram of QUBE-Servo-2 for Section 6.6

As we changed the solver in the Model Settings to the Fixed Step, ode1 (Euler) solver (with step size equal to the sample time of 0.002 used by the HIL blocks), it can be observed there is a noticeable change in the angular speed (See Figures 11 and 12) when compared to the output results using “Auto” in the solver settings.

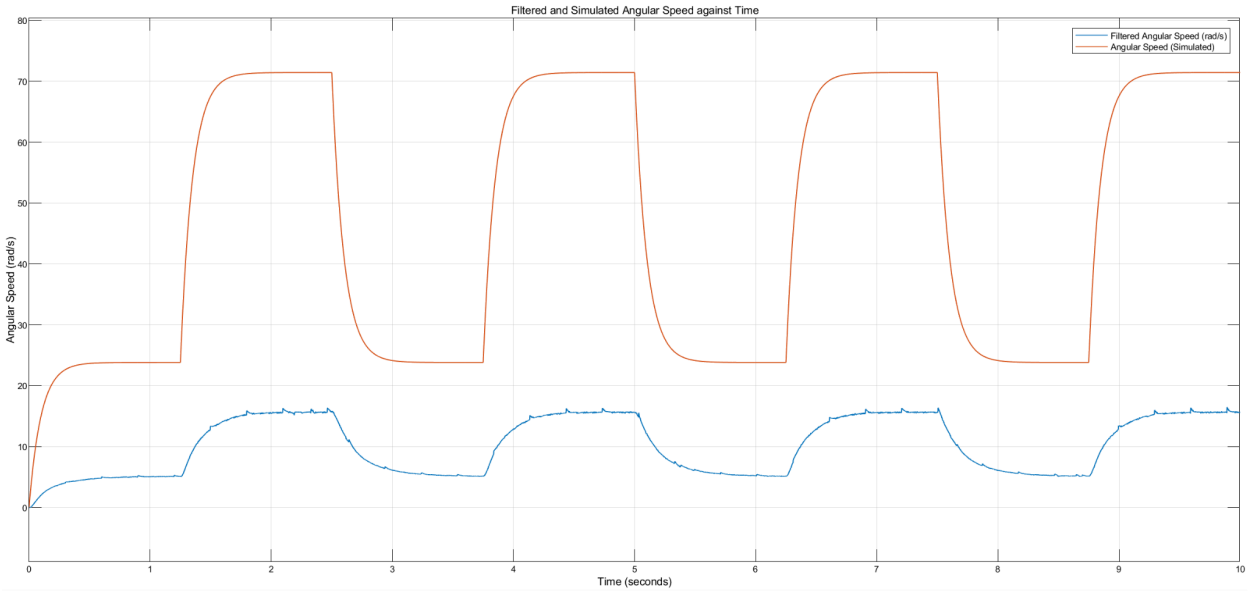


Figure 11: Angular Speed (Filtered Signal and Block Diagram) Against Time using Auto Solver

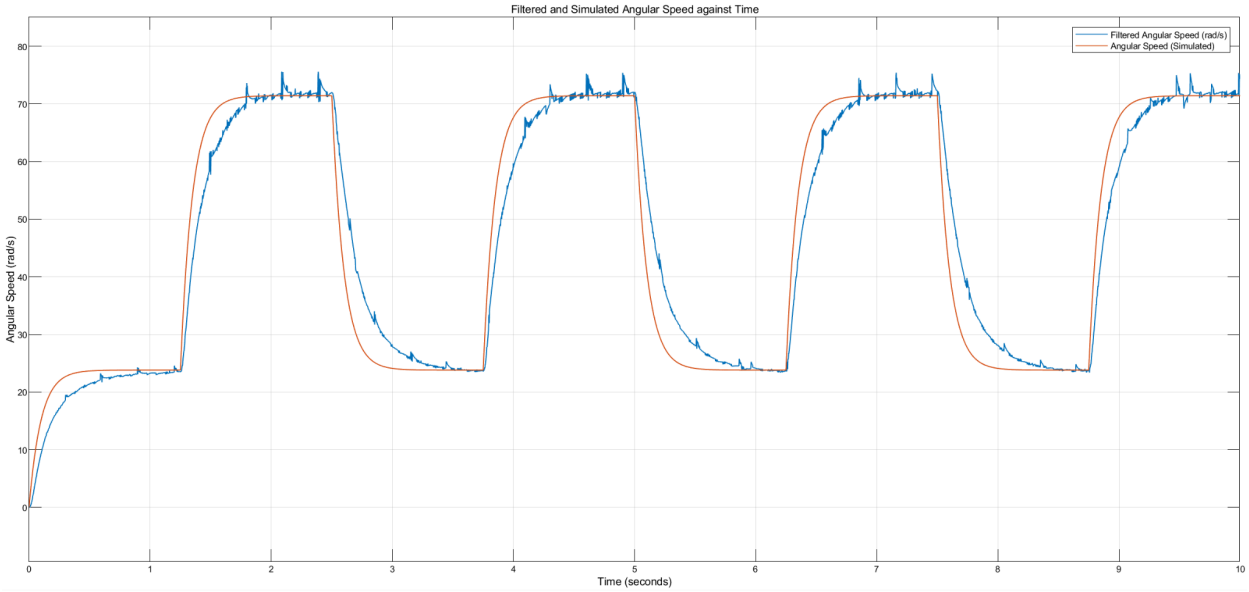


Figure 12: Angular Speed (Filtered Signal and Block Diagram) Against Time using ode1(Euler) Solver

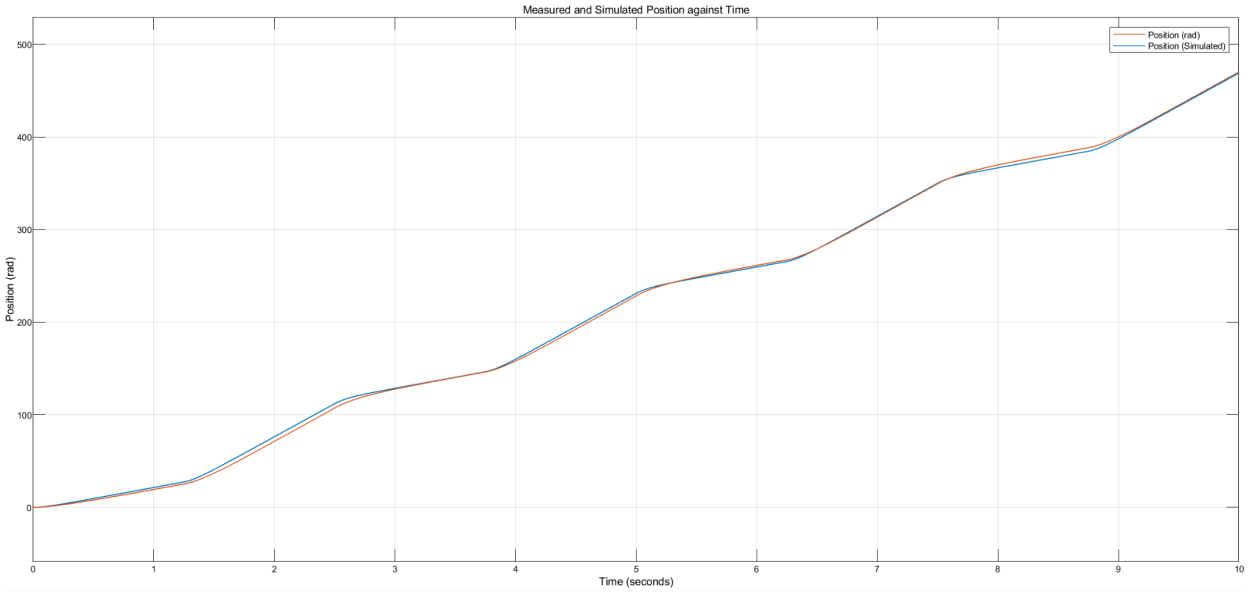


Figure 13: Angular Position (Filtered Signal and Block Diagram) Against Time Plot using Auto solver

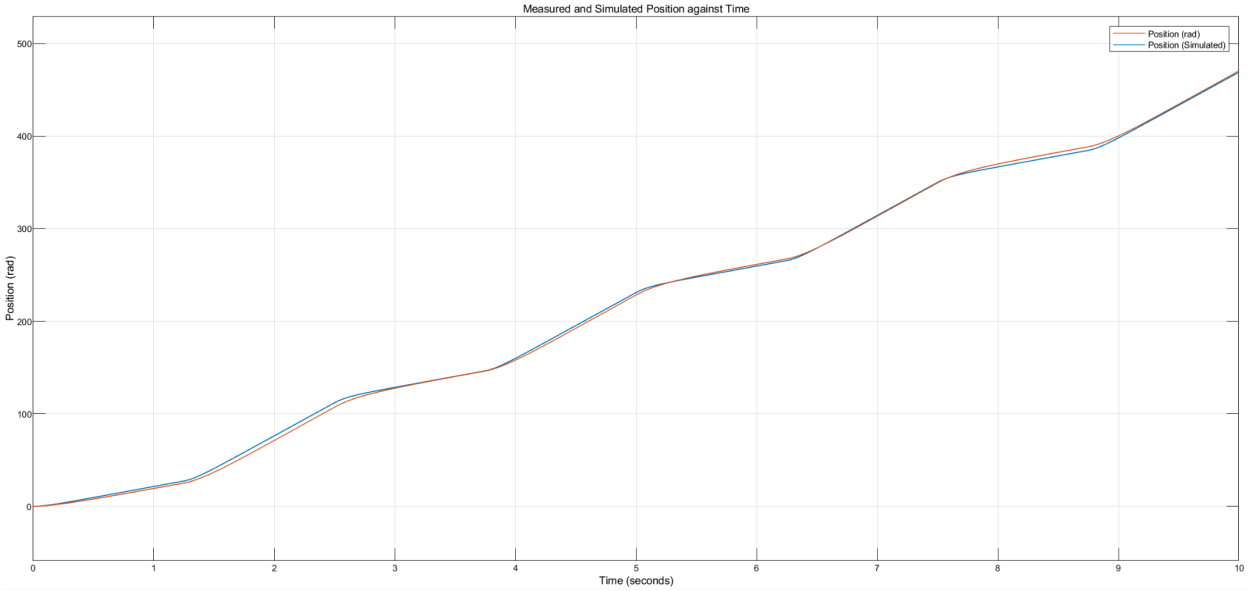


Figure 14: Angular Speed (Filtered Signal and Block Diagram) Against Time Plot using ode1(Euler) solver

6.7: Modelling using Transfer Function: - Simulink Model File (.slx) file attached on D2L

Recall, from Section 6.5, the differential equation:

$$\frac{d\omega_m(t)}{dt} = \frac{k_t}{R_m \times J_{eq}} [v_m(t) - k_m \omega_m(t)]$$

$$\frac{d\omega_m(t)}{dt} = \frac{k_t}{R_m \times J_{eq}} v_m(t) - \frac{k_t \times k_m}{R_m \times J_{eq}} \omega_m(t)$$

Taking the Laplace Transform of this differential equation:

$$s \bullet \Omega_m(s) = \frac{k_t}{R_m \times J_{eq}} V_m(s) - \frac{k_t \times k_m}{R_m \times J_{eq}} \Omega_m(s)$$

We can then determine the transfer function, $\frac{\Omega_m(s)}{V_m(s)}$:

$$\Omega_m(s) \left[s + \frac{k_t \times k_m}{R_m \times J_{eq}}\right] = \frac{k_t}{R_m \times J_{eq}} V_m(s)$$

$$\Omega_m(s) \left[\frac{R_m \times J_{eq}}{k_t} s + k_m\right] = V_m(s)$$

$$T(s) = \frac{\Omega_m(s)}{V_m(s)} = \frac{1}{\left[\frac{R_m \times J_{eq}}{k_t} s + k_m\right]}$$

As we know the numerical values for the variables, we can plug them in:

$$R_m = 8.4\Omega, k_t = 0.042\, N.m/A, \quad k_m = 8.4\, V/(rad/s), J_{eq} = 2.089856 \times 10^{-5}\, kg.m^2$$

$$T(s) = \frac{\Omega_m(s)}{V_m(s)} = \frac{1}{[0.004179712s + 0.042]}$$

Implementation of Transfer Function in Simulink:

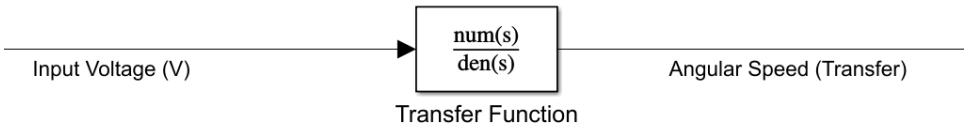


Figure 15: Block Implementation Using a Transfer Function

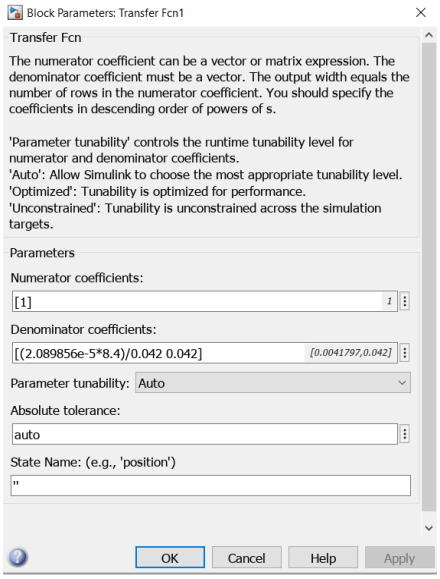


Figure 16: Transfer Function Numerator and Denomerator Values

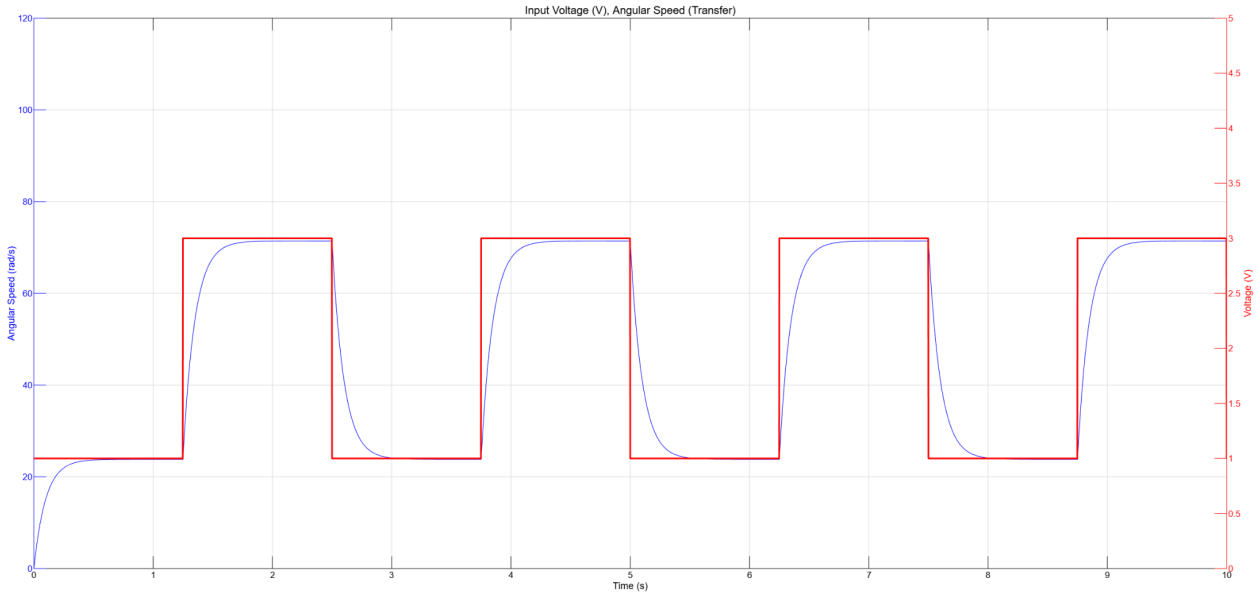


Figure 17: Angular Speed (Transfer) and Input Voltage Against Time

6.8: Comparison:

Plot Comparison:

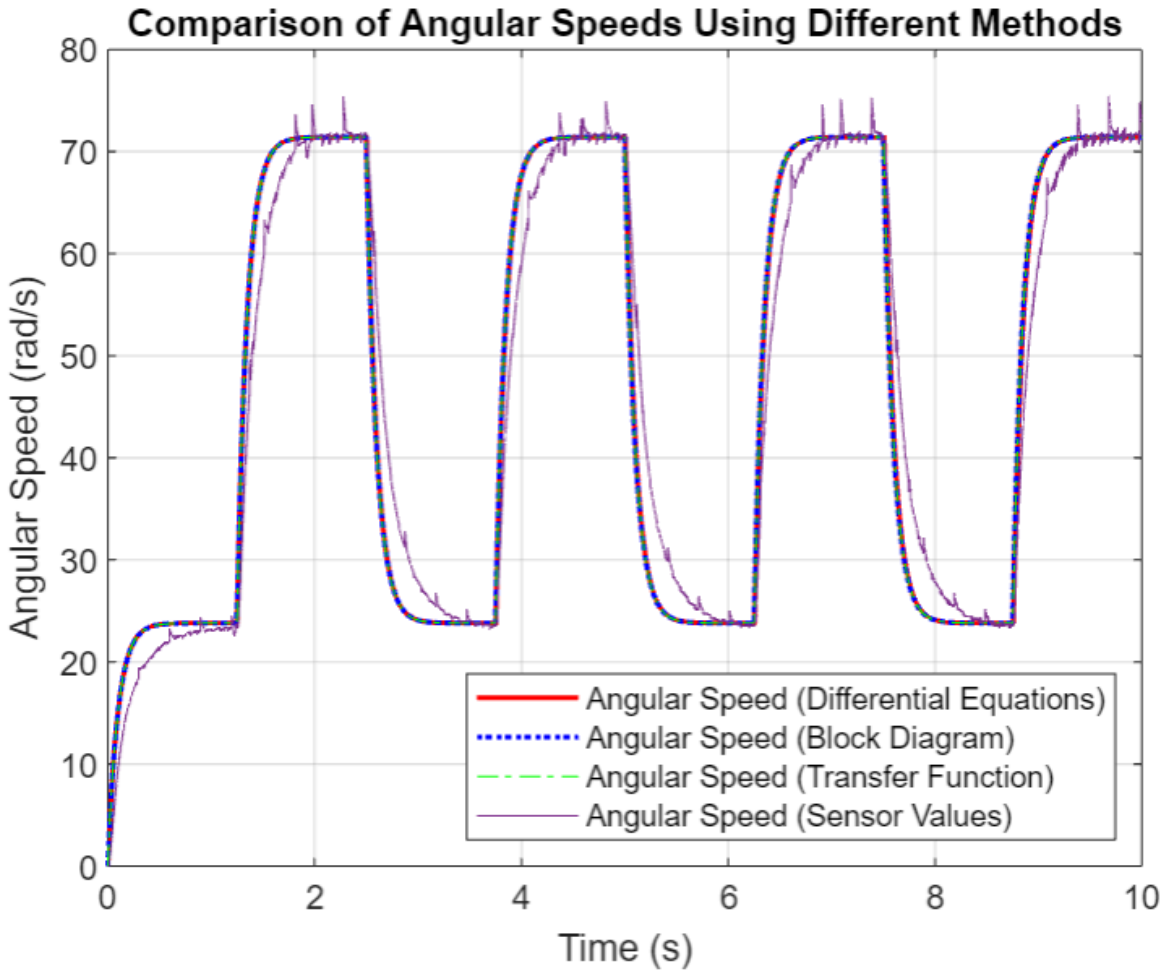


Figure 18: Plot Comparison of all Outputs

The three models give identical outputs because they all solve the same differential equation with the same parameters. The measured sensor output follows the same overall shape and matches the models at steady state, which shows that the models represent the QUBE Servo-2 fairly well. However, it responds more slowly during transients. The noise in the sensor signal angular speed output seen in the plot (small spikes) are caused by the filter's inability to effectively filter out lower frequency noise (< 7.96Hz). and does not cause this lag. One reason the differences are caused by the discrepancy between real and nominal values of R_m and J_{eq}. Another reason is our models used an idealized circuit, where the inductance is ignored. With the experimental value, the inductance is present, which prevents instantaneous changes of current, resulting in the slower transient change in speed from one voltage value to another.